IAM POLICY IN AWS

Introduction

IAM Policy is a core concept in Amazon Web Services (AWS). An IAM policy is a JSON document that defines what actions are allowed or denied for a user, group, or role on AWS resources. In simple words, IAM Policy controls permissions in AWS.

What is IAM Policy?

An IAM Policy is a set of rules that specifies:

- Who can access AWS resources
- What actions they can perform
- Which resources they can access
- Under what conditions

Structure of an IAM Policy

An IAM policy is written in **JSON format** and contains the following main elements:

- **Version** – Policy language version
- **Statement** – Main permission block
- **Effect** – Allow or Deny
- **Action** – What action is allowed/denied
- **Resource** – Which AWS resource

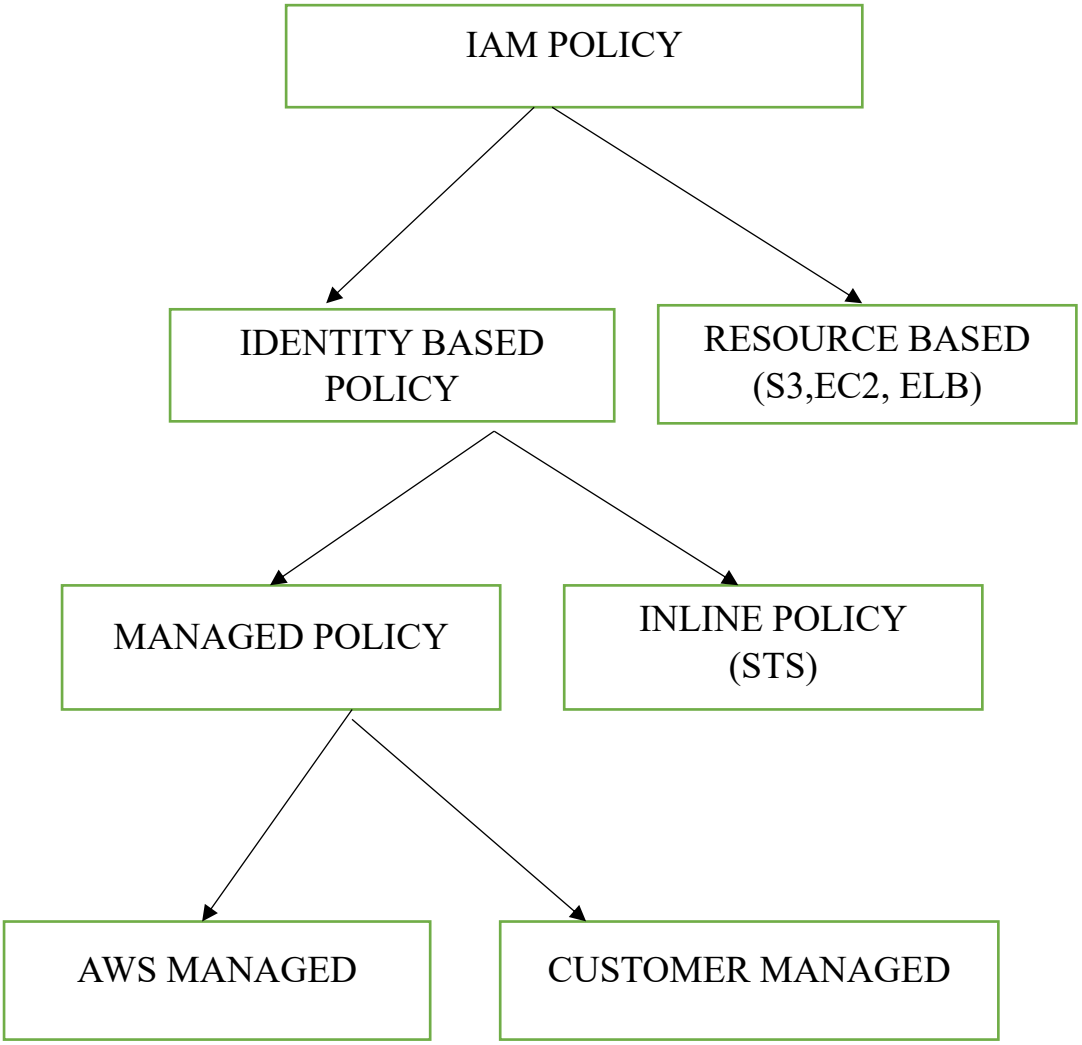
Simple Example of IAM Policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "s3:ListBucket",
      "Resource": "*"
    }
  ]
}
```

Explanation:

- Allows listing of S3 buckets
- Applies to all resources

FLOWCHART



IAM ROLES IN AWS

Introduction

IAM Roles are an important feature of Amazon Web Services (AWS) that allow secure access to AWS resources without using long-term credentials like username and password.
An IAM Role provides temporary permissions to trusted entities such as AWS services, users, or applications.

An **IAM Role** is an AWS identity with:

- A set of permissions (policies)
- No username or password
- Temporary security credentials provided using STS

IAM Roles Uses

IAM Roles are used because:

- They improve security
- No need to store access keys
- Support temporary access
- Enable cross-account access
- Follow best security practices

IAM Role Components

An IAM Role consists of:

1. **Trust Policy** – Defines *who can assume the role*
2. **Permission Policy** – Defines *what actions are allowed*

Two Types of IAM Roles in AWS

1. Service-Based Role
2. Account-Based Role

SERVICE-BASED ROLE (SERVICE-TO-SERVICE COMMUNICATION)

A Service-Based Role is used for communication between two AWS services in Amazon Web Services (AWS).

Explanation

When one AWS service needs to access another AWS service, a service-based IAM role is used to allow this communication securely. Instead of using usernames or access keys, AWS uses IAM Roles with temporary credentials.

Example

- EC2 needs to store data in S3
- An IAM service role is attached to EC2
- EC2 communicates with S3 using the role permissions

Key Points

- Used for service-to-service communication
- No access keys required
- Uses temporary credentials (STS)
-

ACCOUNT-BASED ROLE

An Account-Based Role allows one AWS account to access resources in another AWS account.

Example:

- Account A wants to access S3 in Account B
- Account B creates an account-based role
- Account A assumes the role to access resources

Key Points:

- Used for cross-account access
- Common in large organizations
- Controlled using trust policies
- Improves security between accounts

IAM User vs IAM Role

FEATURE	**IAM User**	**IAM Role**
Credentials	Permanent	Temporary
Username/Password	Yes	No
Best For	Human users	Services & apps
Security	Medium	High

STS (SECURITY TOKEN SERVICE)

STS stands for Security Token Service in Amazon Web Services (AWS).

What is STS?

AWS STS is a service that provides temporary security credentials to users, applications, or AWS services to access AWS resources securely.

STS uses

- To avoid using permanent access keys
- To improve security
- To support IAM Roles
- To allow temporary and limited access

Key Features of STS

- Temporary access keys
- Automatic expiration
- High security
- No long-term credentials

HANDS – ON SCREENSHOTS

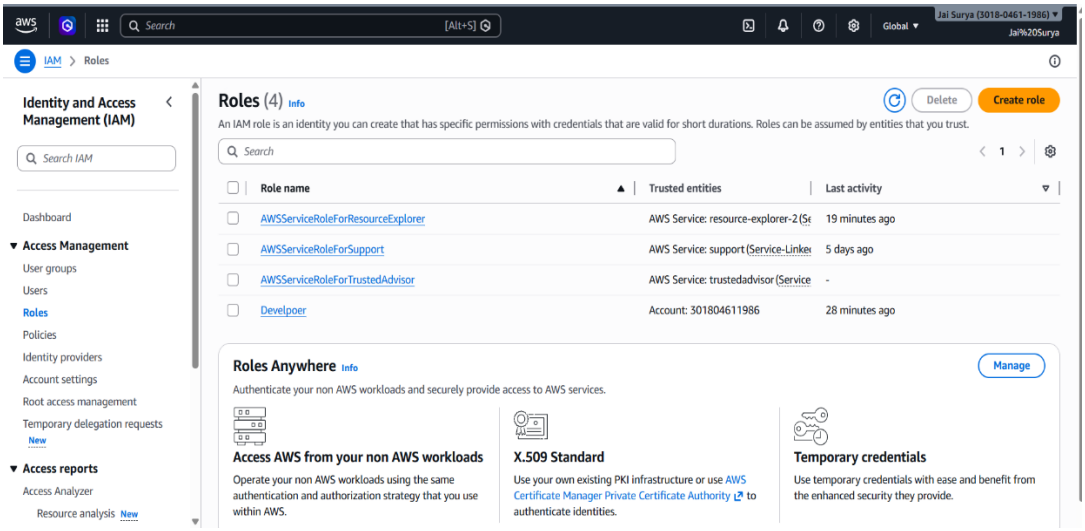
STEP 1: IAM Role Creation

[1] Login to the AWS Management Console

[2] Open **IAM (Identity and Access Management)**

[3] Click **Roles** in the left menu

[4] Click **Create role**



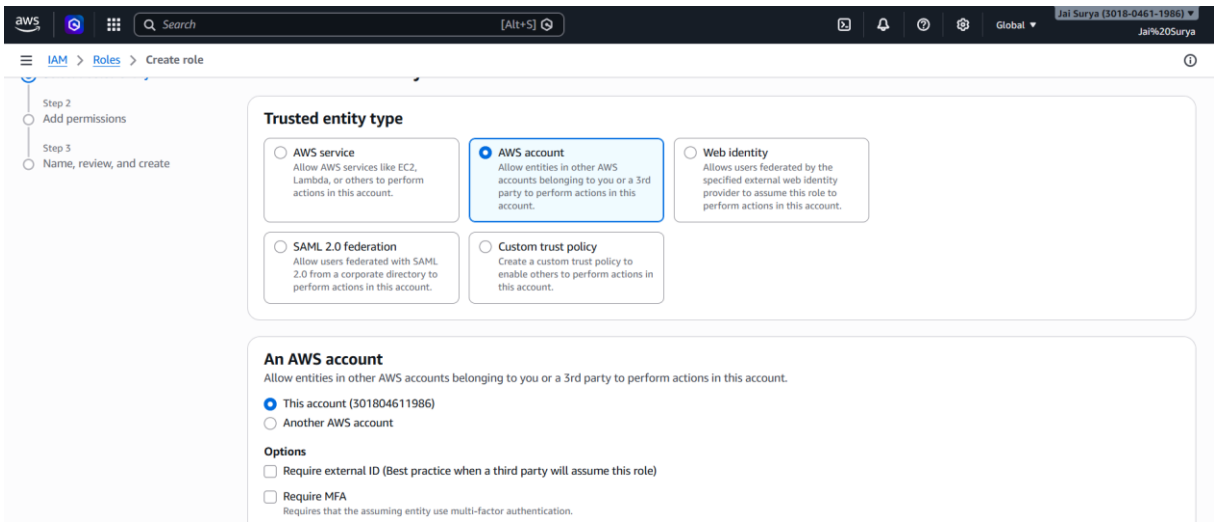
STEP 2: Trusted Entity Selection

[1] In the Create role page, select AWS account

[2] Choose This account

[3] Keep the default options unchanged

[4] Click Next to continue



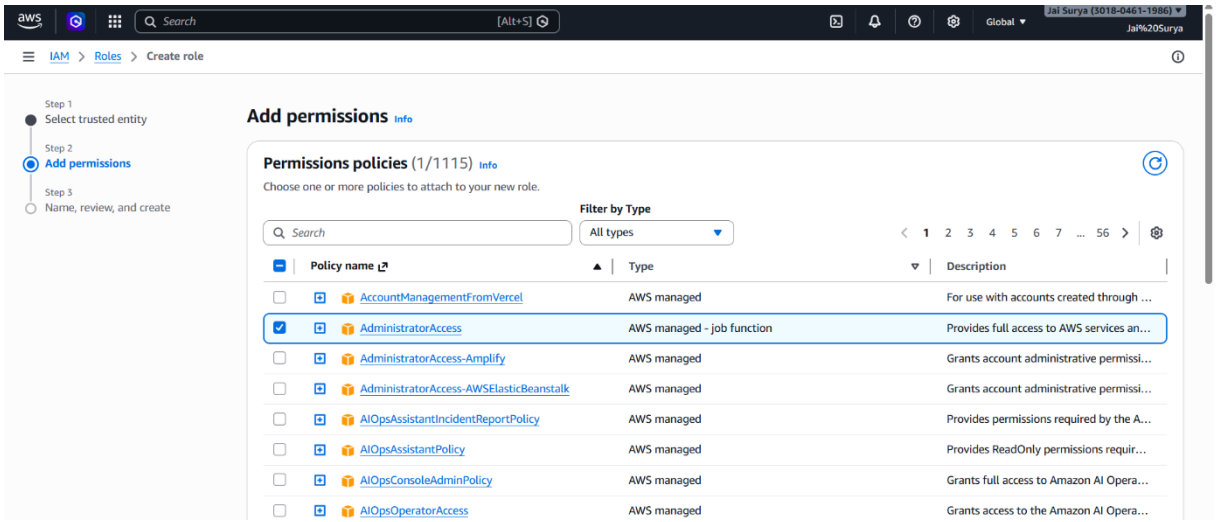
STEP 3: Permission Policy Attachment

[1] In the Attach permissions policies page

[2] Search for the required policy name

[3] Select the policy by checking the checkbox

[4] Click Next to continue



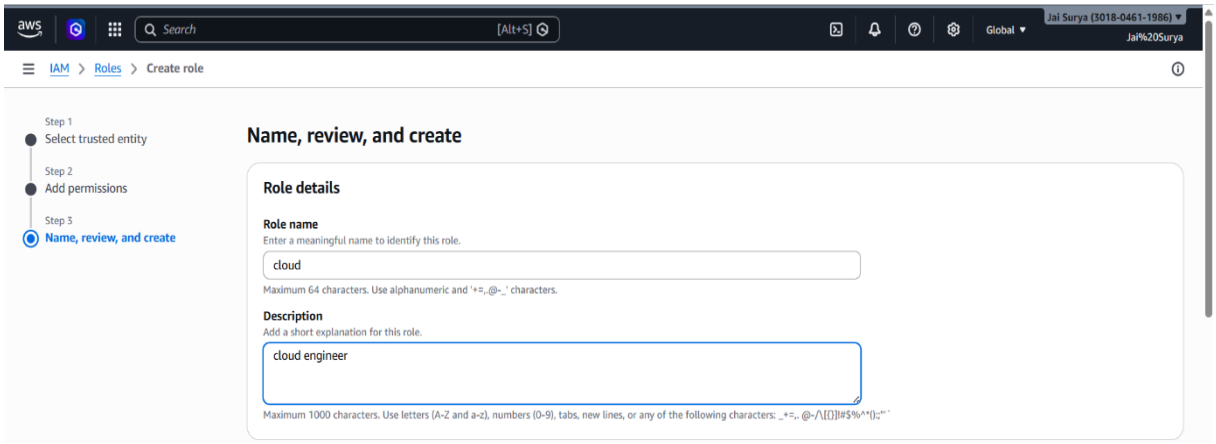
STEP 4: Role Created

[1] Enter a Role name

[2] (Optional) Add a short description

[3] Review the role details

[4] Click Create role



STEP 5: IAM User Creation

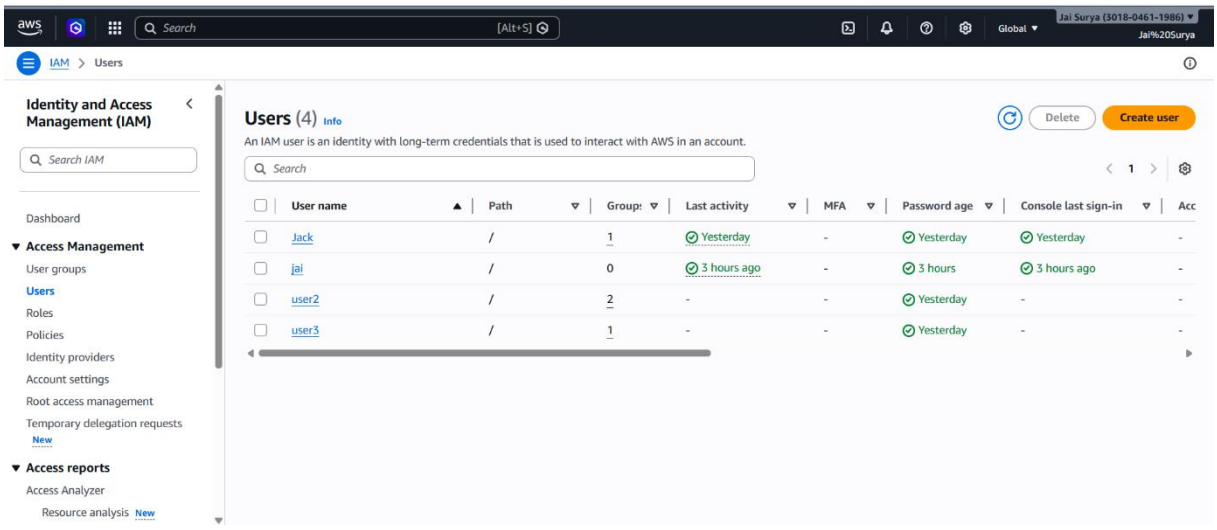
[1] Go to Users in IAM

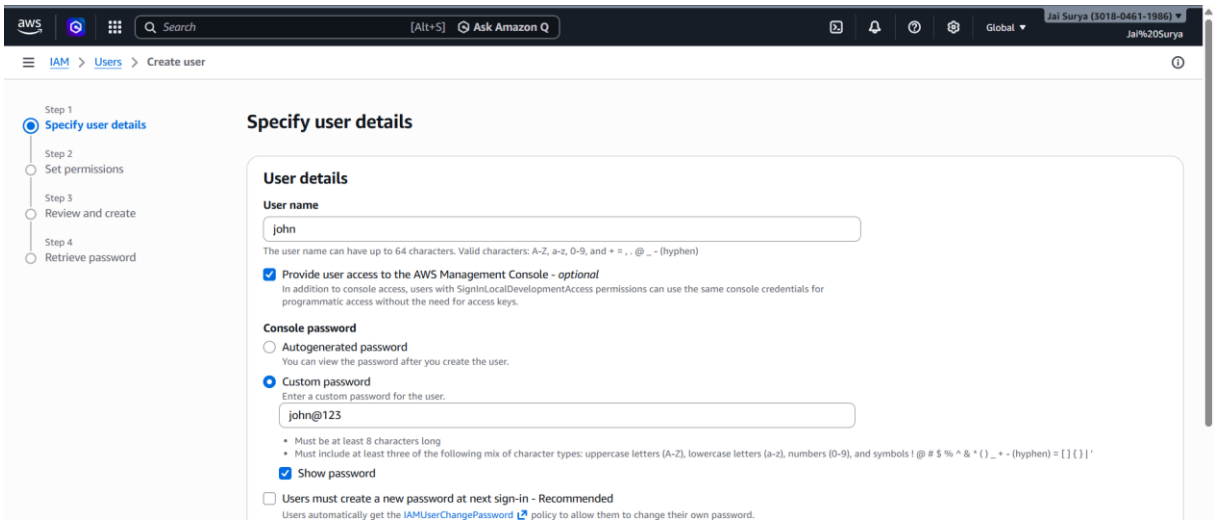
[2] Click Create user

[3] Enter the User name

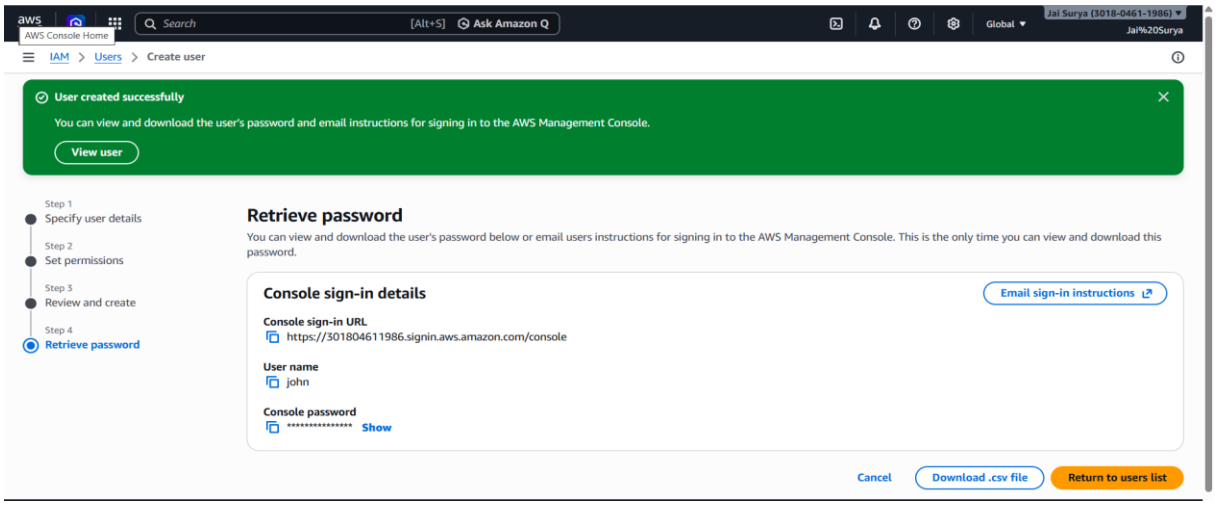
[4] Select AWS Management Console access

[5] Click Next





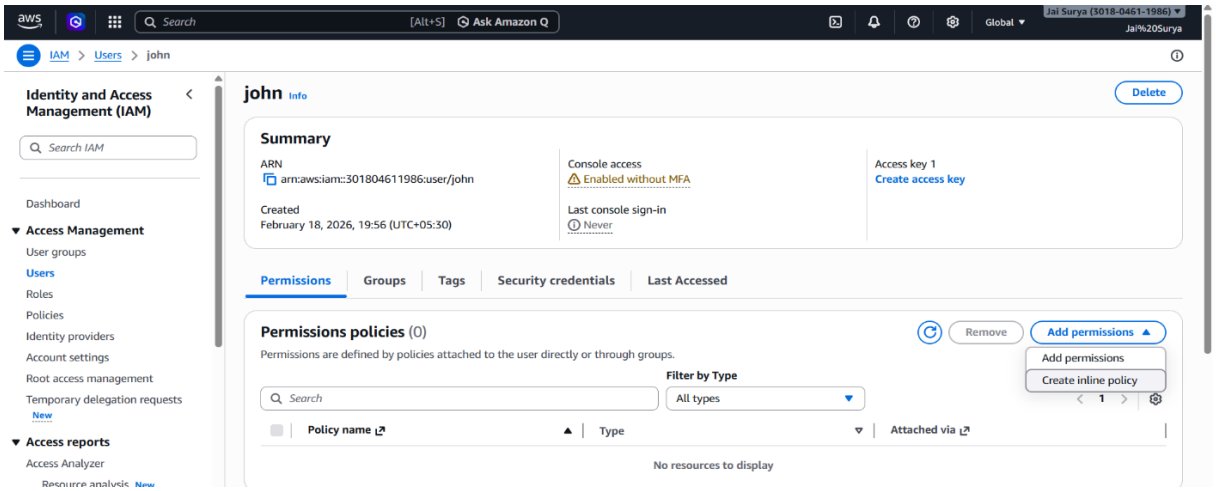
[1] User created successfully



STEP 6: Inline Policy

[1] Open the Permissions page for the user

[2] Select Add inline policy



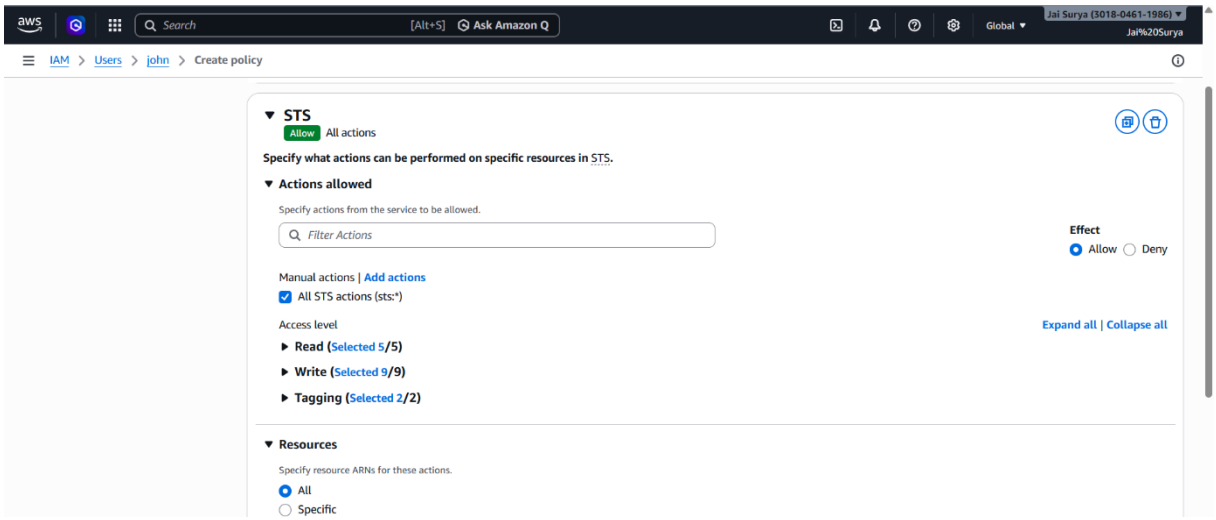
STEP 7:STS Policy

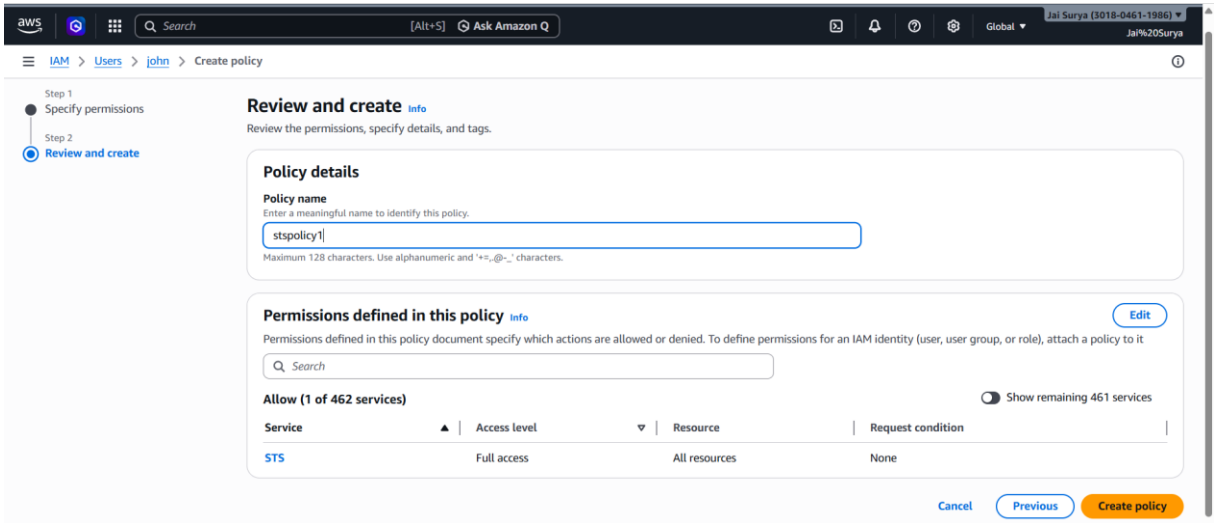
[1] Add sts: AssumeRole permission

[2] Review the policy

[3] Enter policy name

[4] Click Create policy





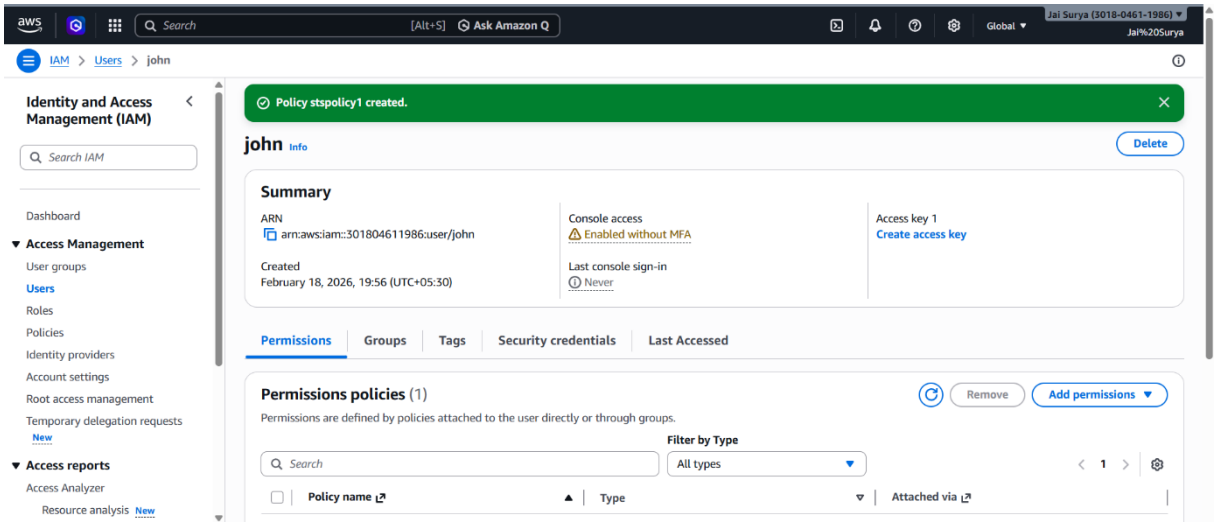
STEP 8: Policy Created

[1] Inline policy created

[2] Policy attached to the IAM user

[3] User now has STS Assume Role permission

[4] Policy Created Successfully



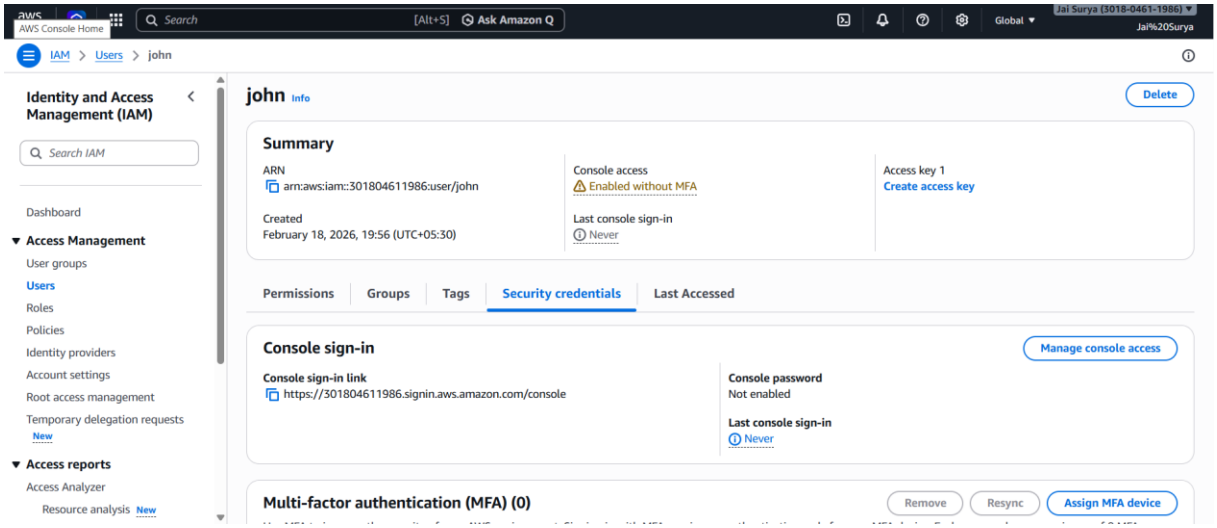
STEP 9: IAM User Login

[1] Open the IAM user

[2] Go to Security credentials tab

[3] Scroll to Console sign-in link

[4] Copy the console login URL



STEP 10: IAM User Login

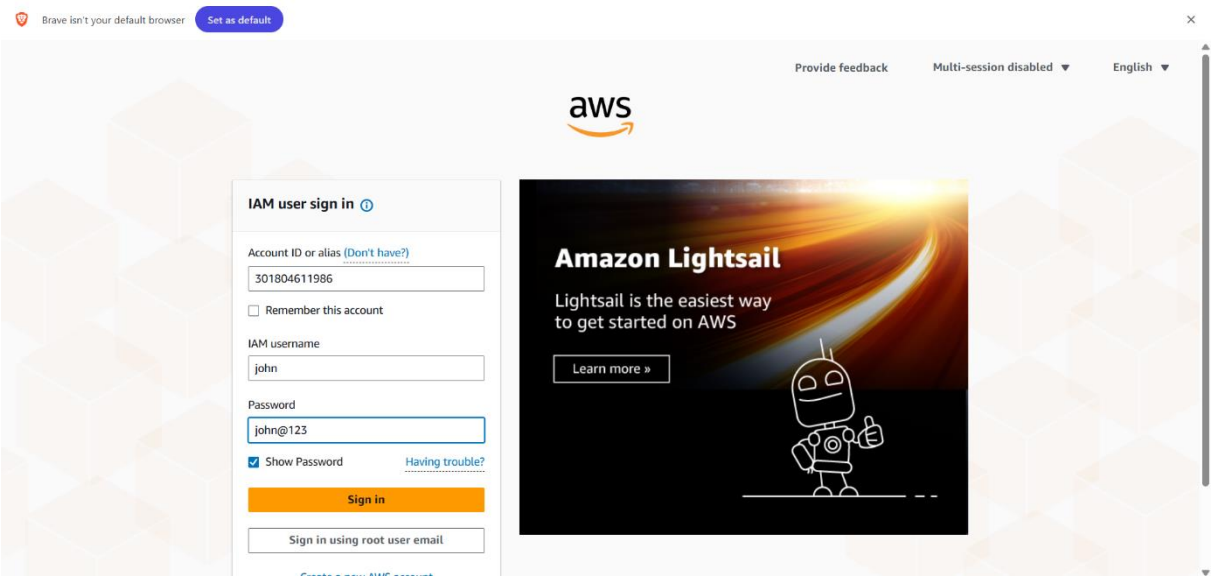
[1] Paste the console login link in a new browser tab

[2] Enter the Account ID

[3] Enter IAM user name

[4] Enter password

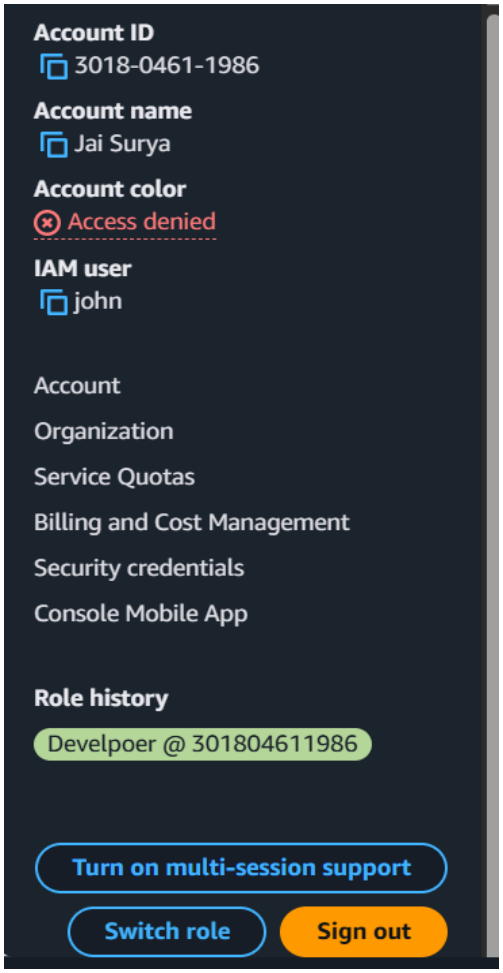
[5] Click Sign in



STEP 11: Switch Role

[1] Click the profile icon (top-right corner)

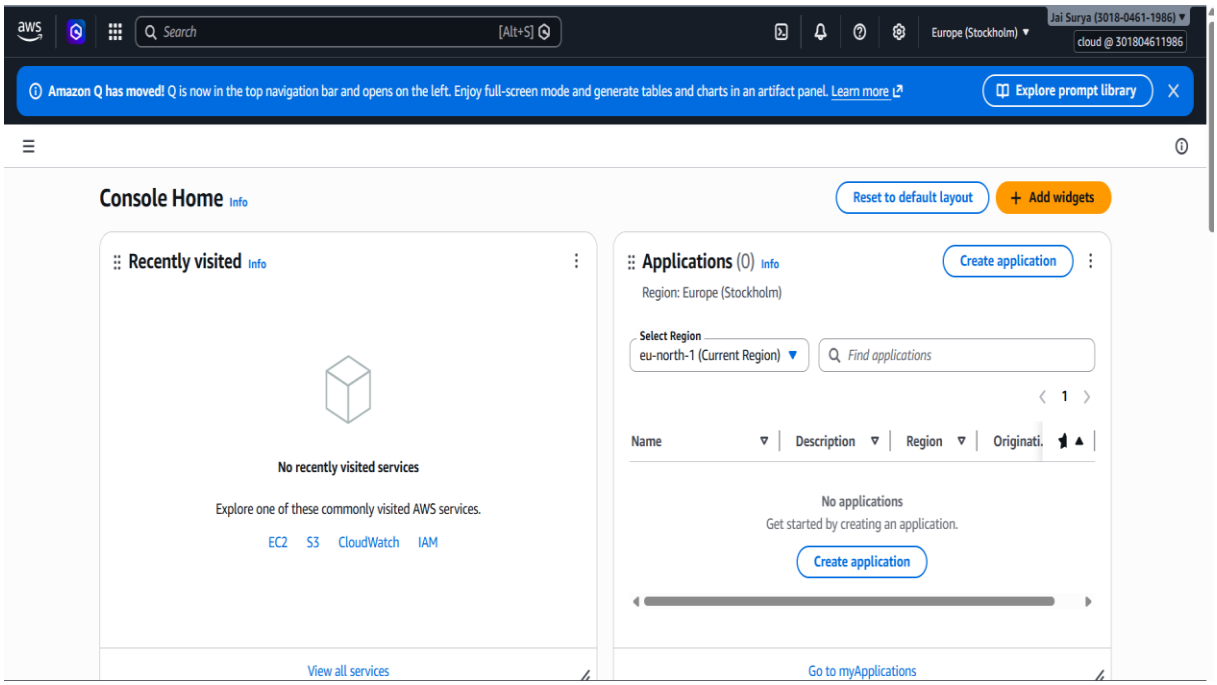
[2] Select Switch role from the menu



STEP 12: Role Switch Success

[1] Enter Account ID

[2] Enter Role name

[3] Click Switch Role

[4] Role switched successfully